

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2.5: Regression

Jana Lasser

Forschungsgruppe

Complex Social & Computational Systems (CS)²

IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz



Supervised Learning

- Up to now, we have only seen how to predict **classes** using supervised learning.
- However, we can also predict (continuous) **numbers**.
- We still use the supervised learning approach (with example data for training the model), but instead of classification, we perform **regression**.

Supervised Learning: Regression

- A number of *observations* with *attributes* and their *values* $(x_1, \dots, x_d)$.
- For each observation, a *continuous* target value $y \in \mathbb{R}$
→ here: speed.

Goal: Find the correct *target value* for a new observation.

→ Regression

Example

Speed = 68 km/h *Speed* = 58 km/h



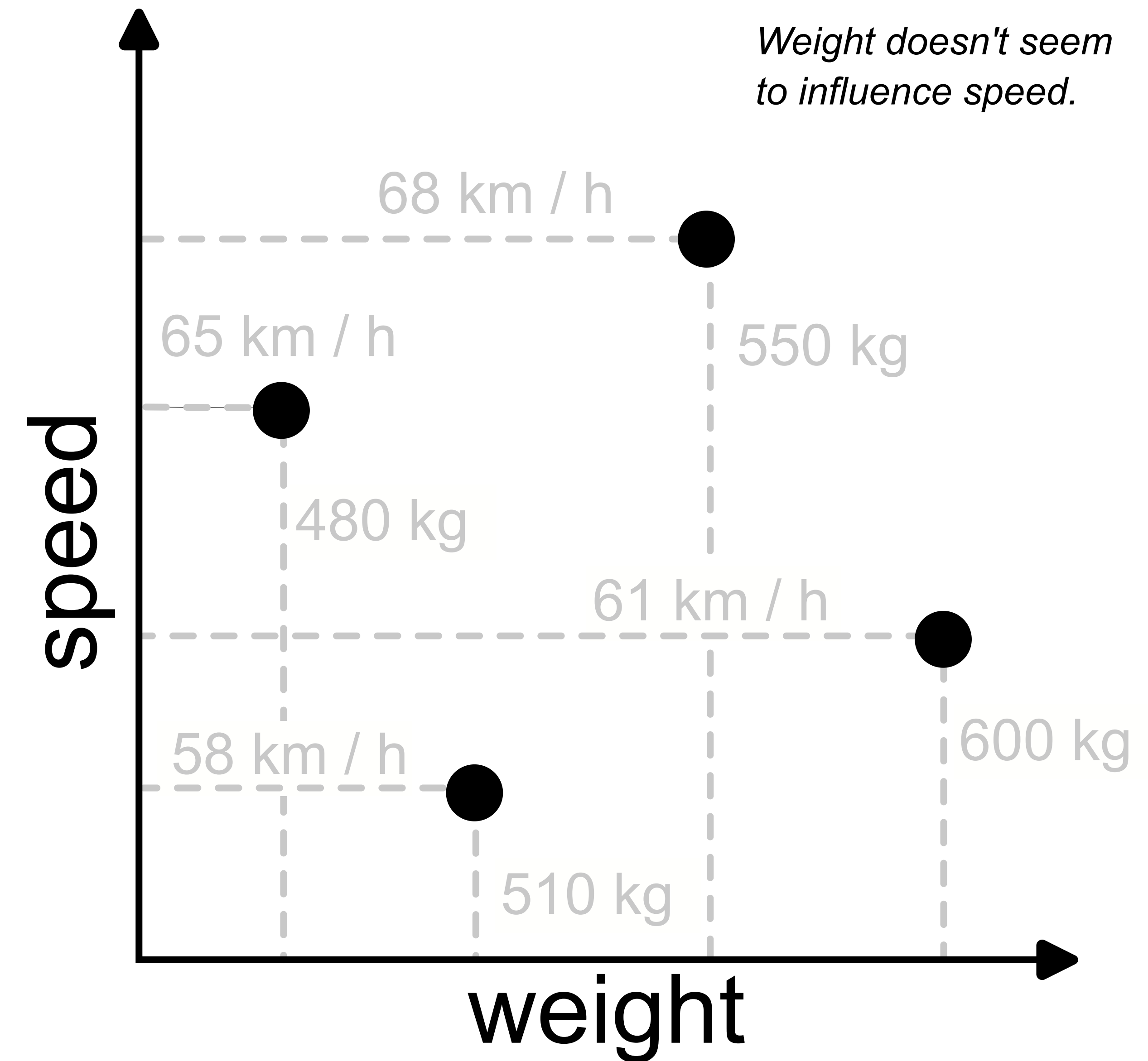
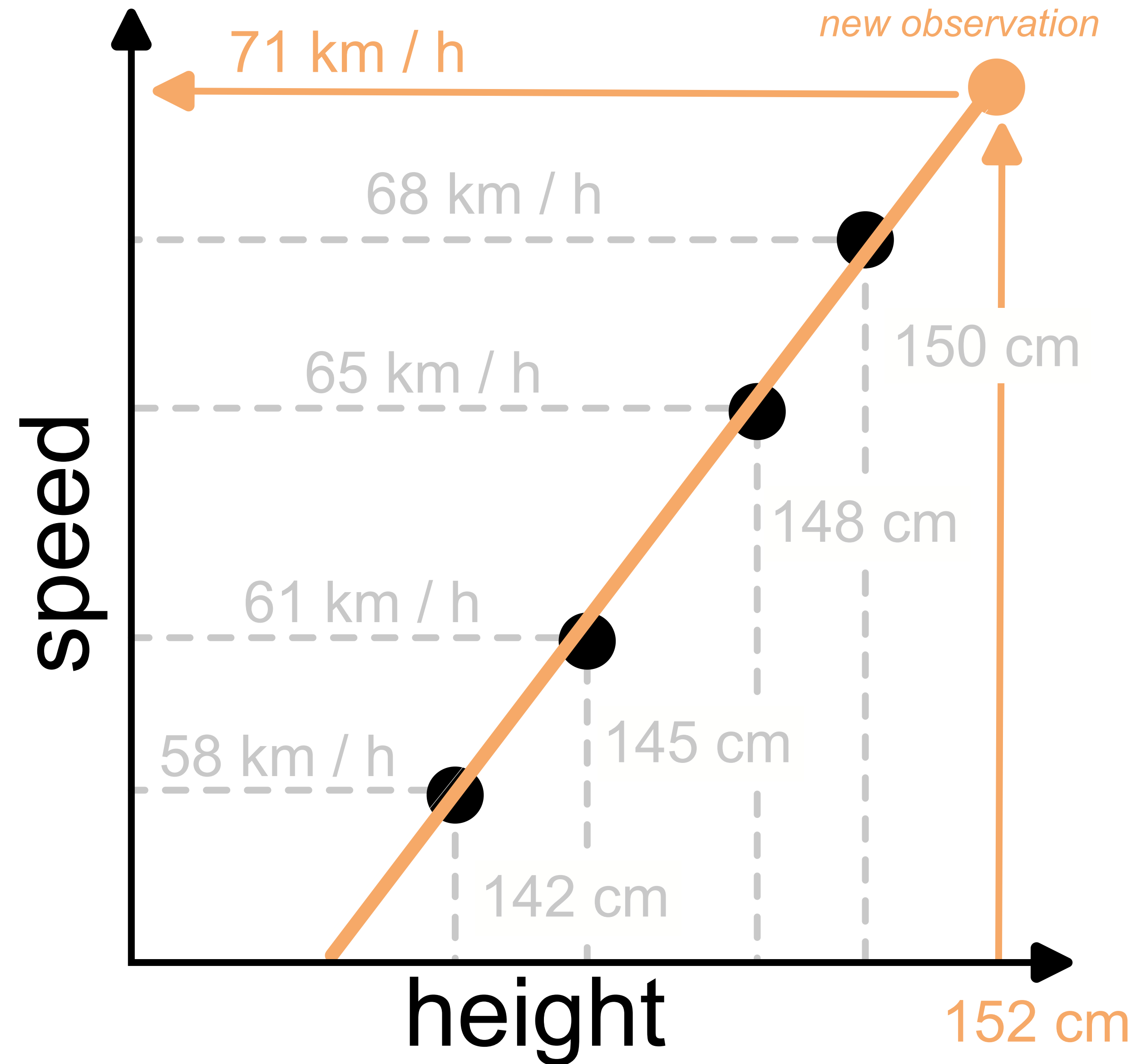
Speed = 61 km/h *Speed* = 65 km/h



Observation: Horse weighing 520 kg and 148 cm tall. → predicted speed: 66 km/h

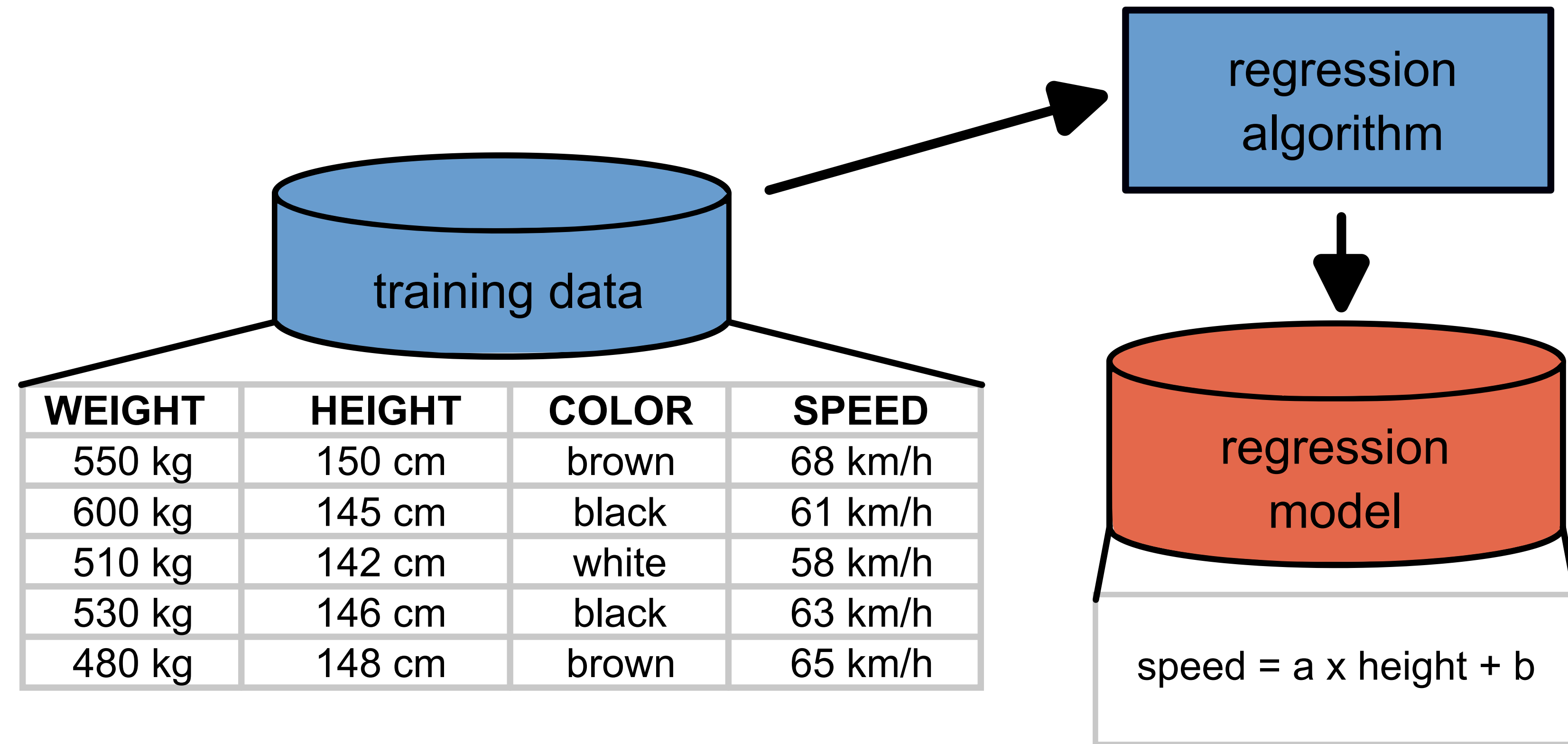
Supervised Learning: Regression

Find a line that passes through all points.

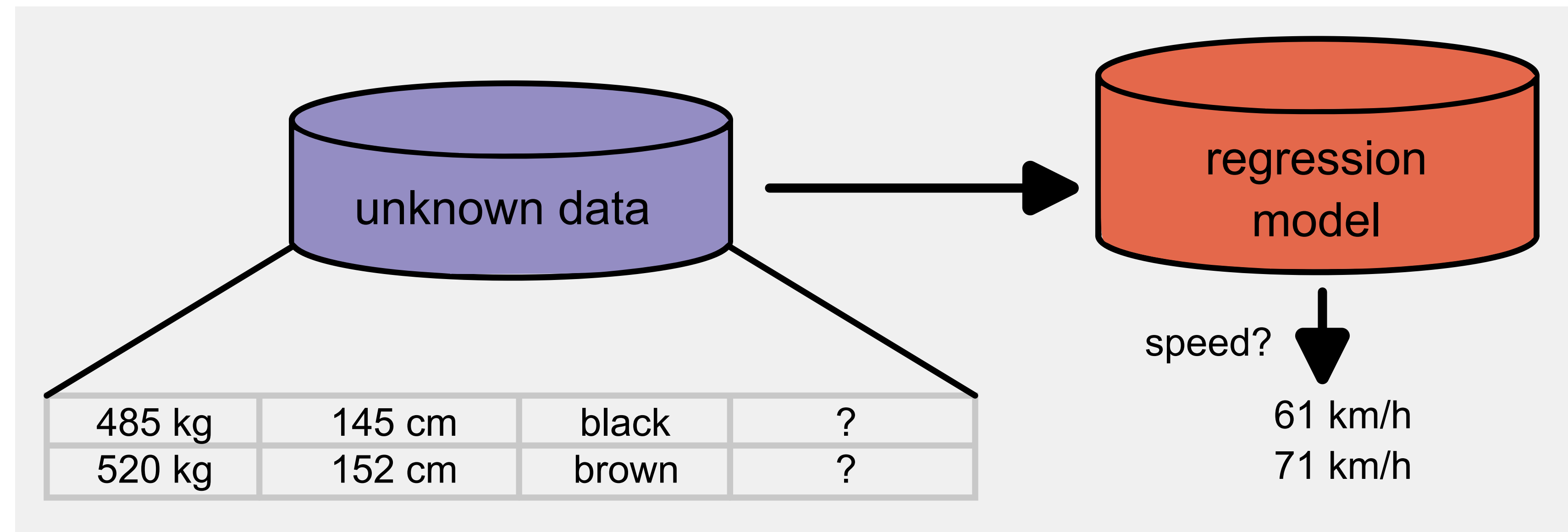


Regression "Pipeline"

Training Phase

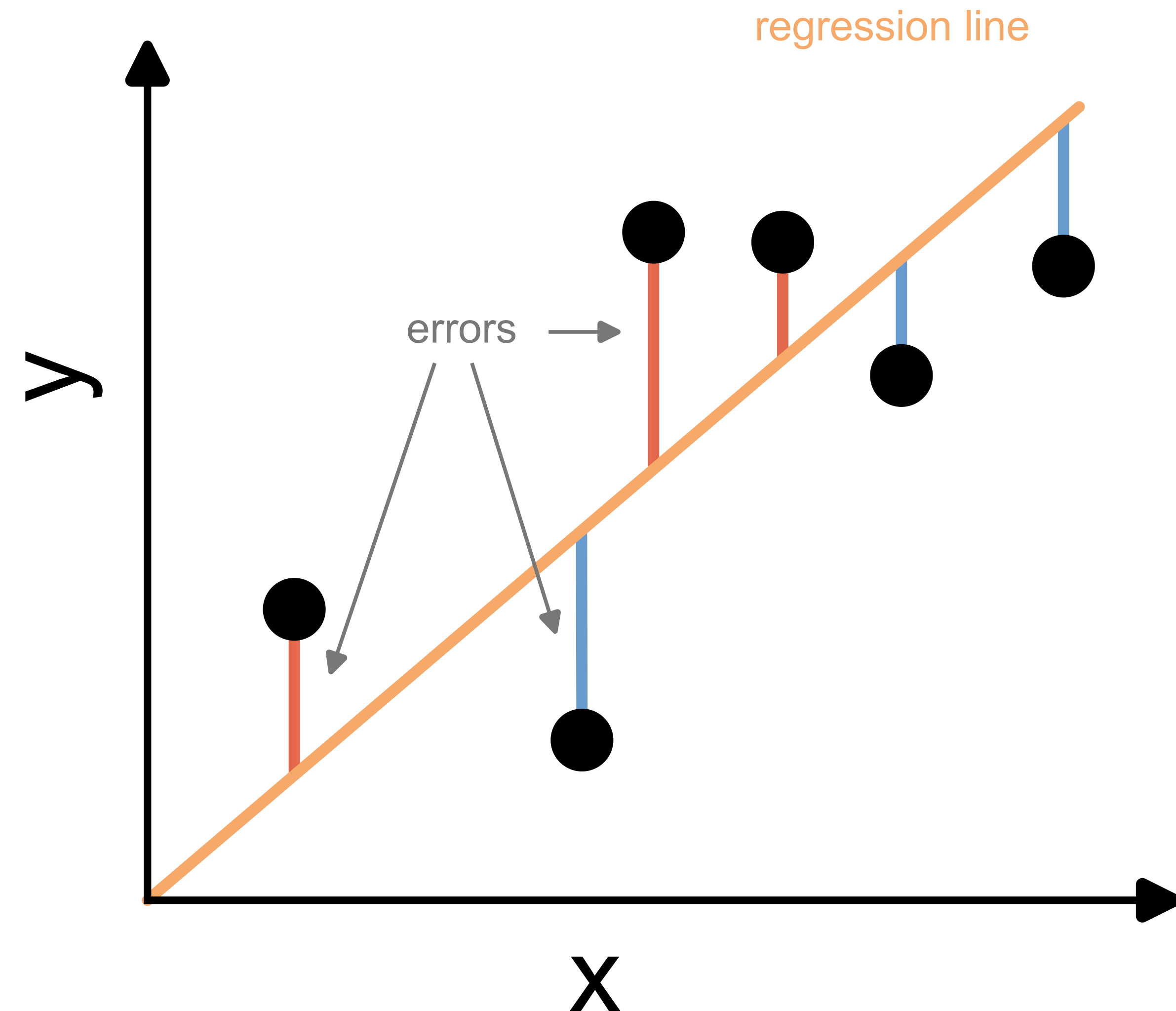


Application Phase



Mean Squared Error

Since we are not predicting discrete classes, we cannot use error types like false positive or false negative. → we need different error measures.



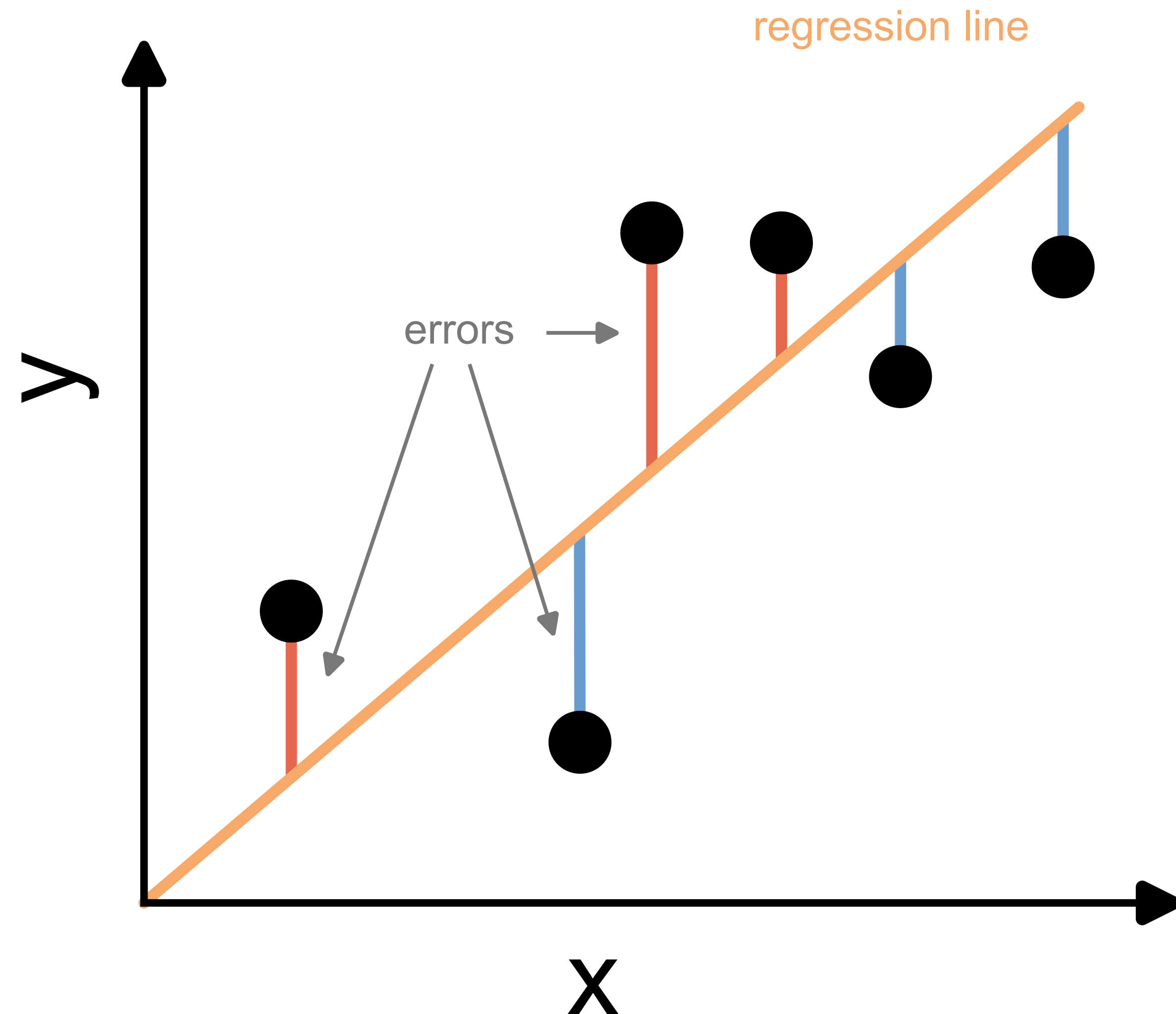
mean error:

square root
of the mean
squared
errors

Diagram illustrating the calculation of the standard deviation of the sample mean. It shows two sets of vertical bars representing data points. The first set has three bars with heights +9, +4, and +1, each preceded by a '+' sign. The second set has two bars with heights +4 and +1, each preceded by a '+' sign. To the right of the second set is the equation: $= \sqrt{28/6} = 2.2$.

Mean Squared Error

The square root of the mean squared error is also called "root mean squared error" (RMSE).



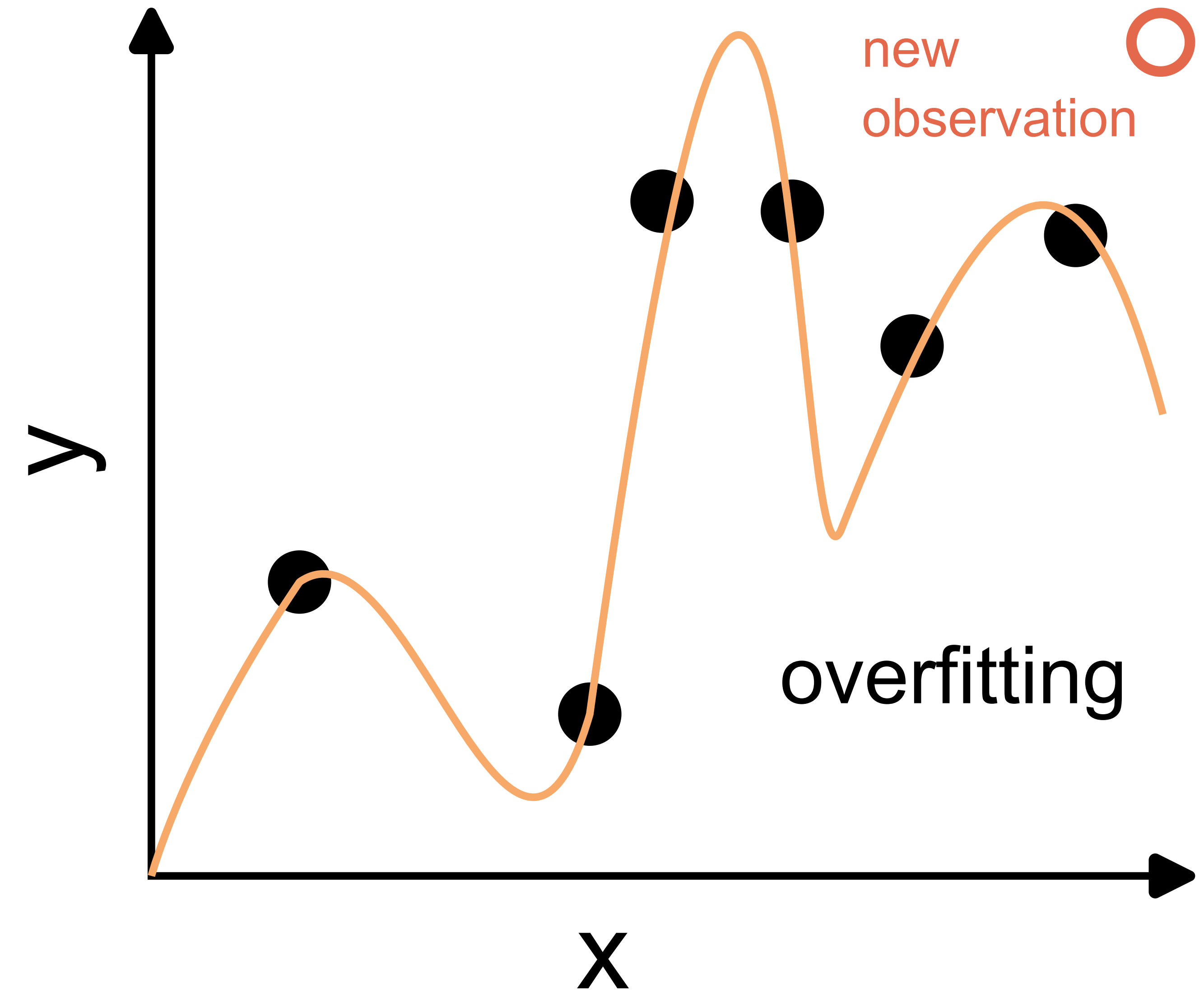
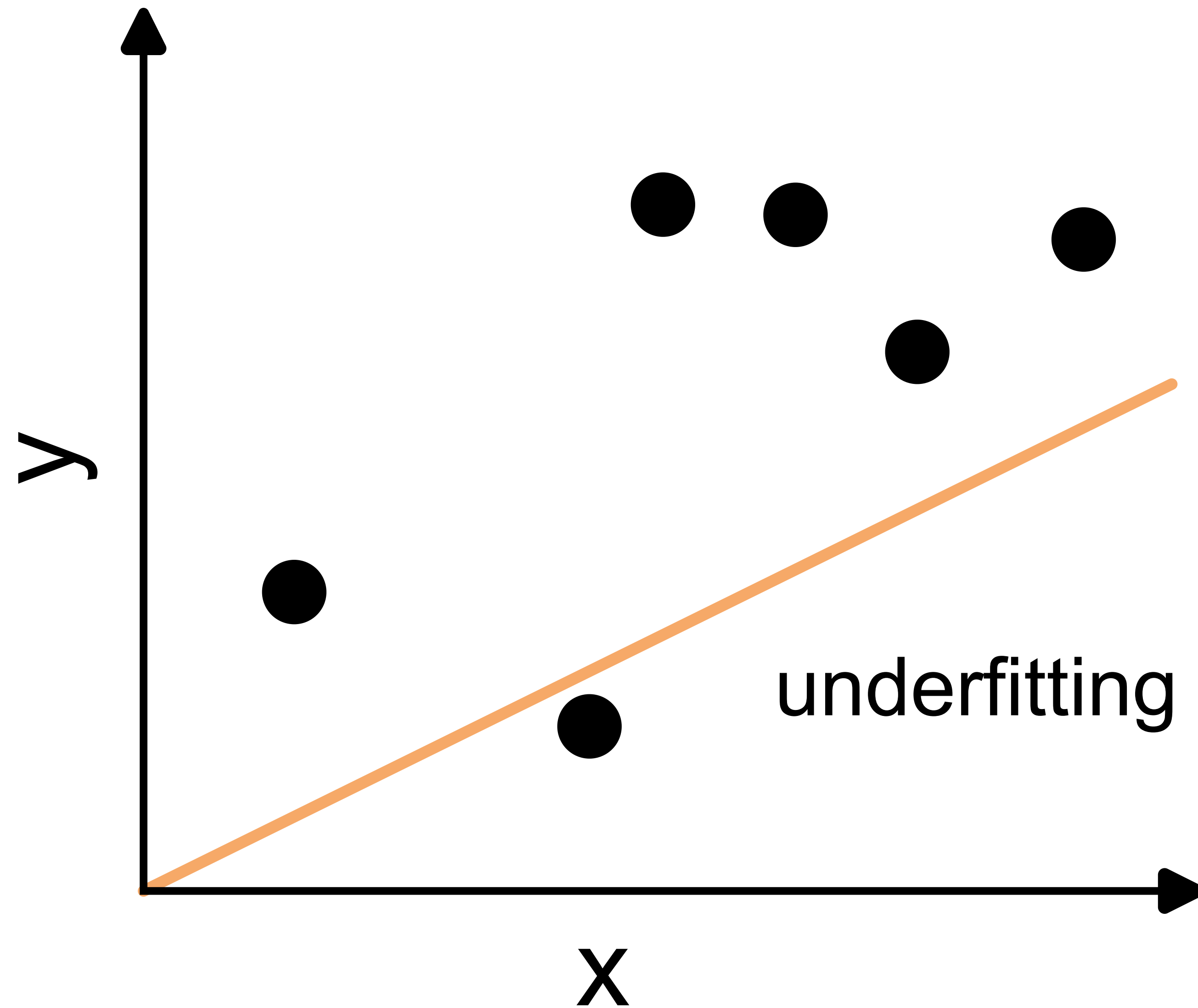
1. By squaring, it does not matter whether errors go up (positive) or down (negative).
2. Due to the squaring, large errors have a higher weight than small errors and can be "penalized" more heavily.

Alternative:

Pearson correlation coefficient

Bias and Variance of a Regression

Bias (underfitting) and Variance (overfitting) are very intuitive to understand for regressions.



Summary

- Supervised learning can be used for both **classification** (predicting classes) and **regression** (predicting numbers).
- Regression follows the same training-prediction logic as classification.
- As a measure for the performance of a regression, we use the root of the mean squared error.